Construction of Smart City and Resilient City from the Perspective of the Metropolitan Area Coupling Analysis

Hanyuan Li[1(✉)], Lishuang Jia[2], Jiannong Liu[1], and Chenxi Ma[3]

[1] Department of Engineering Management, Sichuan College of Architectural Technology, Sichuan 618000, China
524163934@qq.com
[2] School of Public Administration, Southwest Jiaotong University, Sichuan 610000, China
[3] School of Architecture, Southeast University, Nanjing 210096, China

Abstract. This paper based on the data of Deyang City from 2018–2022, by using TOPSIS to evaluate comprehensive level of construction of resilient city and smart city in Deyang city to find the coupling coordination between the two. In order to achieve a higher level of mutual coordination and mutual facilitation between the two systems, a coupling coordination model of the two systems was built, and then analyzing the obstacle factors. The results show that, the comprehensive level of construction of Resilient City and Smart City in Deyang city shows an upward trend, and the coupling degree between the two has been relatively good, and the coupling coordination degree is also increasing year by year. The coupled coordination between the two systems changed from Mild-obstacle to High-quality-coordination. Through the analysis, the main subsystems are Ecological Environment and Smart Energy, and the main obstacle factors are unemployment rate, comprehensive production capacity of water supply, energy consumption intensity, etc., four countermeasures are put forward from actively promoting college education reform, promoting practical teaching, and ensuring the stable completion of students, etc..

Keywords: Coupling Coordination · Resilient City · Smart City

1 Introduction

Over the past few years, the urban environment has been impacted by many natural disasters and epidemic diseases. Resilient cities, as one of the concepts from urban planning, has become the latest trend in urban construction. In April 2020, President Xi, highlighted at general secretary of the central financial committee seventh meeting, "build livable city, resilient city, smart city, build high quality urban ecosystem and security system." The party's 20th report also emphasizes that "to update the national security infrastructure and capabilities, firmly protect national security and social order" and "safeguard new pattern of development with a new security pattern." The thorough implementation of the regional coordinated development strategy is the objective need

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025
B. He et al. (Eds.): UDCC 2024, LNCE 729, pp. 3–30, 2025.
https://doi.org/10.1007/978-981-95-1573-8_1

to build a high-quality territorial spatial layout, but also the inevitable requirement to achieve high-quality development [1]. Building a resilient, livable and smart city has become an important development direction of urban construction and renewal, and is a major strategic deployment for urban work in the new era and new stage [2]. Sichuan province attaches great importance to resilient city work, *Sichuan national territory spatial planning (2021–2035)*, emphasized the ecological priority, saving priority, put forward to promote harmonious coexistence between man and nature, strongly pushed to build livable toughness efficient urban space.

Resilience city is a "multi-in-one" system in which multiple elements promote each other and the benefits overlap. Promoting the construction of resilient cities and optimizing the comprehensive urban disaster prevention planning can maintain the stability of cities and help cities to build an environment for harmonious development. However, how to deal with major disaster risks, bring larger regions such as urban and rural areas, urban agglomerations, metropolitan areas into the research scope, and build a regional integrated comprehensive risk prevention system, is a practical problem in the current comprehensive disaster prevention planning practice [3]. A comprehensive database of smart and resilience indicators can assist city managers and relevant departments complete the of resilient cities [4]. With the progress of smart infrastructure construction, the construction of smart city constantly has been improving, and its perfect smart city service facilities can also promote the construction of urban resilience. The construction of smart cities helps to create a resilient society in a crisis, and the integration of smart city energy systems as an Energy Hub (EH) aids in addressing the difficulties faced by the power sector amidst the pandemic and achieves a significant degree of robustness [5].

Therefore, it is significant to promote coordination analysis of urban resilience and intelligent construction. At present, Deyang city is pushing forward the construction of an important node of Deyang Chengdu-Chongqing metropolitan economic circle, and making every effort to build a world-class major equipment manufacturing research and development base. It is necessary to promote and improve the environmental safety and disaster prevention and recovery capacity of Deyang city. So, Deyang city can be an good example for coupling analysis of construction for the smart city and resilient city. It is of positive significance to use the intelligent construction of the city and maintain the resilience of the city to build it into a world-class major equipment manufacturing research and development base. Specifically speaking:

Theoretically speaking, it will expand the research field of the coordinated development of Chengdu-Chongqing twin economic circle, enrich the connotation of coordination between smart city and resilient city, and help improve the application of urban smart infrastructure in the process of resilient city construction.

Practically speaking, it will provide a theoretical basis for the government regulatory departments to improve the urban infrastructure construction, provide feasible and operable measures for building smart cities and resilient cities, and explore the sustainable promotion of the construction and development of a healthy city in the Chengdu-Chongqing metropolitan area.

2 Research Methodology

2.1 Research in Related Fields

Academics have carried out a lot of research on the ecological transformation of infrastructure and the implementation of urban resilience. In the field of infrastructure construction, Many scholars analyze the ecological resilience and sustainability of urban green infrastructure by constructing ecological evaluation indicators [6]. In order to form a very good indicator system, after collecting government laws and regulations, government and expert research reports, various almanac books and authoritative survey data, detailed analysis of research hotspots is beginning from key words of Domestic and foreign literature. From CNKI, web of science (WOS), Scopus, Engineering Index (EI) databases, searching the keywords of "resilient city" and "smart city", the first 1,000 articles were selected for the analysis. In the end, CNKI 1000, WOS 1000, Scopus 1000 and EI 1000 articles were selected in different databases.

Statist ical Table of Keywords Research on Resilient Cities

Keyword Nunber		Centrality	Keyword Strength	
Urban Resilience	124	0.56	Disaster Prevention and Mitigation	2.12
Resilience	54	0.19	Resilient Transportation	1.51
Econonic Resilience	51	0.15	Flood Resilience	1.21
Digital Economy Resilient Governance	33	0.1	Vulnerability	1.21
	28	0.05	Sponge City	1.03
Coupling Coordination	20	0.02	Knowledge graph	0.9
Resilience evaluation	19	0.03	Old residentia area	0.9
Influence Factor	18	0.07	Psychological resilience	0.9
Urban Agglomeration	17	0.08	Traffic planning	0.9
Resilience Assessnent	15	0.12	Resilient Comunity	0.84

Smart City Keyword Research Statistics Table

Keyword	Strenpth	Keyword		Centrality
Digital Twin	1.52	Digital Econony	58	0.06
Smartification	1.35	Digital Twin	32	0.07
Indoor Navigation	1.18	Internet of Things	28	0.02
Target	1.18	Big Data	25	0.08
Public Service	1.18	Artificial Intelligence	25	0.07
Network Socurity	1.18	Urban Governance	22	0.03
Enterprise Innovation	1.18	Smart Coumunity	20	0.03
Snart Streetlights	1.14	Digital Government	16	0.02
Social Governance	0.91	Digital Governance	14	0.01
Data Sharing	0.91	Metaverse	13	0

Fig. 1. Summary chart of keyword analysis data of the research literature on resilient cities and smart cities in China. (Self-drawn)

Domestic keyword analysis of resilient city and smart city, as shown in Fig. 1. From Deyang native perspective, scholars like Liao Hanxiang [7, 8] study Deyang resilient city evaluation, build the evaluation index system and concluded that Deyang city resilience is in medium level, and put forward some number of relevant management measures. Besides, Deyang has achieved good results in the city brain data sharing, cross-sectoral business coordination, urban fine governance, especially great achievements are finished in the garden city, key technology, ecological construction. Construction path as "six in one", "cloud brain fusion", "four network operation" is highly promoted, which give an big empowerment to city smart operation, and Deyang walked out of a "Ecological Smart City" construction mode, which is of great significance to Deyang smart city

construction [9–11]. From the perspective of the country, some evaluation indicators, such as highway problems and rainwater accumulation are put forward [12, 13]. Huang Jinzhou [14–16] put forward smart facilities such as smart transportation and the Internet of Things will play an important role in the future urban construction, and the application of artificial intelligence in this area should be strengthened.

From the international point of view, based on resilient city, the detection, prediction and management platform of urban rain flood prevention and control is an important part of resilient cities [17–19]. As for the smart cities, it is necessary to promote the application of the Internet of Things, smart city platform and digital twin technologies in cities through policy support and communication and cooperation among relevant stakeholders [20, 21].

The keyword data of Scopus and EI are summarized below (Figs. 2, 3,4 and 5).

From the above, it can be easily seen that economic level, transportation, policies and regulations, smart infrastructure, etc., some articles also give some reference indicators. But, based on native and foreign literature analysis, it can be seen that resilient city and smart city research has certain cross, but the evaluation standard is not very unified, and there are little researches on coupling of resilient city and smart city. In addition, intelligent means are mostly used for the reform of urban transportation and energy, in the process of selecting the evaluation indicators of smart city and resilient city in Deyang city, the indicators that can be directly borrowed are still limited. Therefore, in the process of choosing indicators, indicators from field of energy, economic production and transportation mentioned above should be fully considered. However, through the above analysis, it can be easily seen that the construction of smart city and resilient city in Deyang city also has unique requirements, such as the natural green space environment, disaster response ability, etc. Ecological environment can be judged by green coverage rate, air quality and other factors. In addition to disaster response ability, as mentioned in relevant literature, can be measured by economic indicators, Deyang city highly emphasizes disaster prevention ability, so water supply, road and medical treatment can also be included as disaster response indicators.

In this study, more thinking should be combined with the urban positioning of Deyang. The basic ideas and research direction of relevant research can be used for reference and become the main reference of the evaluation index of system model, but they cannot be copied. The specific index should be combined with the local situation of Deyang City.

2.2 Modeling and Evaluation Methods

Coupling originates from a concept in physics, which is a measure of the degree of interdependence between two entities in physics [22]. A city is a huge and complex system, and the construction of resilient cities and smart cities is becoming more and more perfect. Whether the coordination relationship and interaction between the two city systems are benign needs to be carefully studied, and can be used to evaluate this model. There are many models to measure correlation relationships, such as functional relationships (linear, nonlinear), correlation relationships, and so on. But in dealing with the problems of two systems, but the coupled coordination model can reflect the degree of coordination between the systems and reflect the stage of the level of coordination

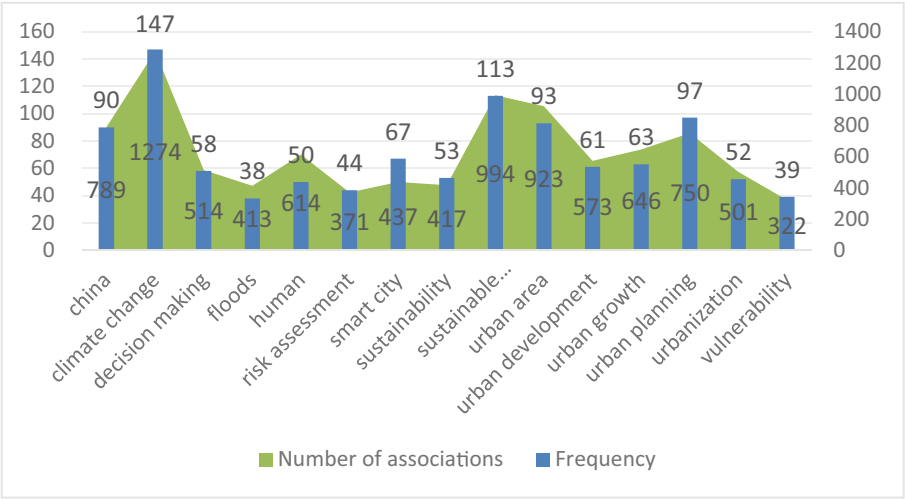


Fig. 2. Map of the frequency and number of related keywords "resilient city" from Scopus. (Self-drawn)

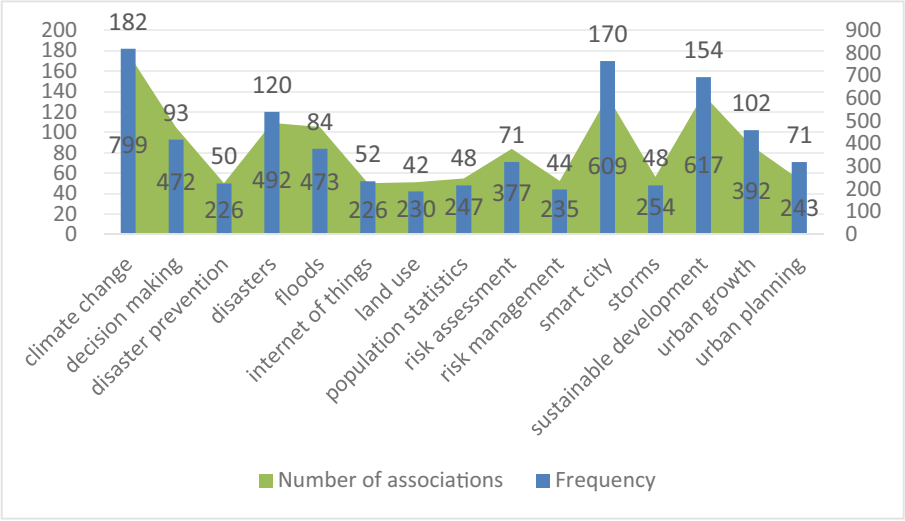


Fig. 3. Map of the frequency and number of related keywords "resilient city" from EI. (Self-drawn)

development, which is suitable for the coordination for Smart City and Resilient City from a city. Since this study has only two systems, so the resilient system is R_1, smart system is R_2, the specific formula is as follows:

$$C = 2\sqrt{f(R_1)f(R_2)\Big/\left(f(R_1)+f(R_2)\right)} \tag{1}$$

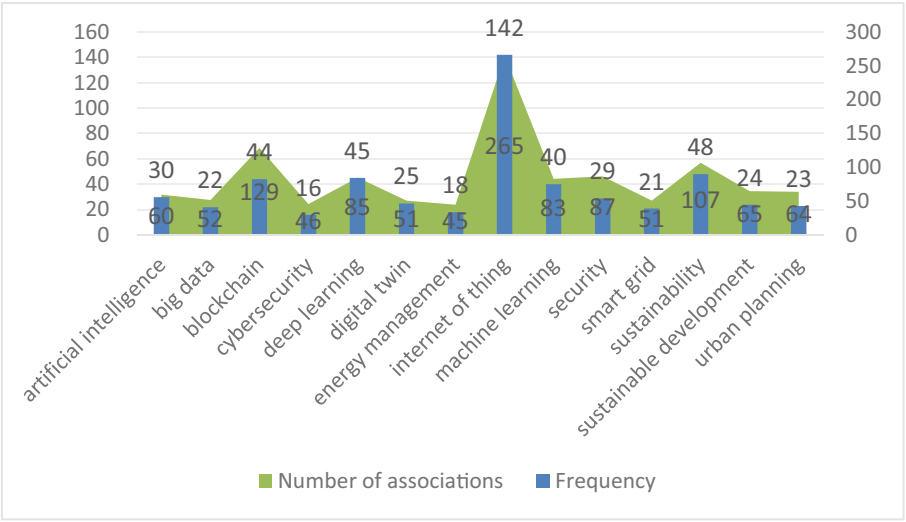


Fig. 4. Map of the frequency and number of related keywords "smart city" from Scopus. (Self-drawn)

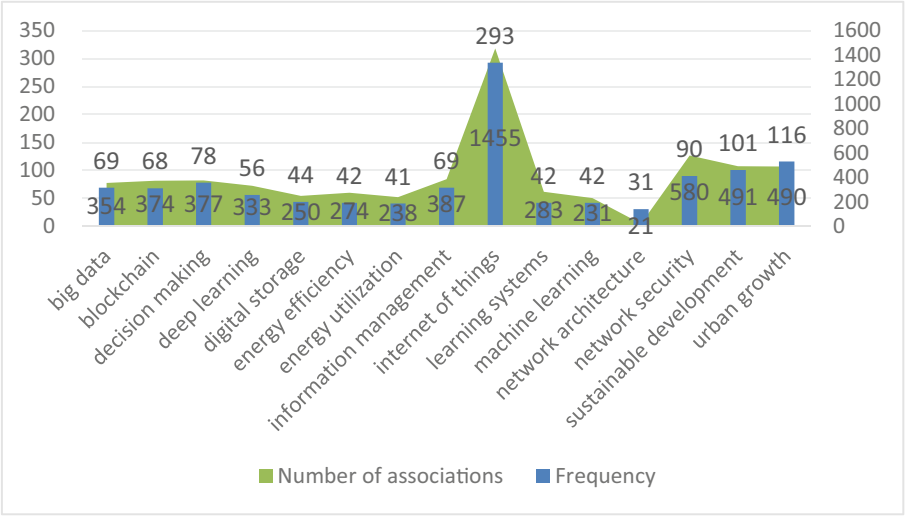


Fig. 5. Map of the frequency and number of related keywords "smart city" from EI. (Self-drawn)

$$\mathrm{T} = \alpha f(R_1) + \beta f(R_2) \tag{2}$$

$$\mathrm{D} = \sqrt{\mathrm{CT}} \tag{3}$$

In the formula, Cis stand for the coupling degree between the two systems, $f(R_1)$ and $f(R_2)$ are the comprehensive evaluation index of the two systems, respectively. The

coupling degree between the two systems can be judged by the size of C. And T is a comprehensive development index, α is the contribution share to resilient urban systems, β is contribution share to the smart city system, D is the coupling coordination degree of the two systems. In fact, the construction and development of smart cities and resilient cities are both important directions of urban territorial spatial planning and national construction, In relevant studies, they are both highlighted. So, in this research, $\alpha = \beta = 0.5$. As for the level of the coupling degree, different experts have different evaluation classification. For example, For example, Miao Chenglin [23] and Zhang L [24] classify 5 levels for the coupling degree of urbanization and the ecological environment. In order to avoid complicated work in research and considering the large amount of research factors in this project, considering the analysis has a lot of data of economics and digitization, five levels (Table 1) are selected for reference [25].

Table 1. Standard table of levels for coupling degree and coupling coordination degree between resilient cities and smart cities. (Self-drawn)

Coupling	Level	Coupling coordination	Level
[0.0, 0.2)	Low-level	[0.0, 0.2)	Extreme-obstacle
[0.2, 0.4)	Mid-level	[0.2, 0.4)	Mild-obstacle
[0.4, 0.6)	Run-in stage	[0.4, 0.6)	Basic-coordination
[0.6, 0.8)	High-level	[0.6, 0.8)	Good-coordination
[0.8, 1.0]	Benign-trend	[0.8, 1.0]	High-quality-coordination

With many economic and production data collected in this research, the comprehensive evaluation of the two systems will use the entropy-TOPSIS method. In the traditional TOPSIS method, each index is given weight through subjective evaluation, which is usually difficult to judge the importance of each indicator, and it is more appropriate for experienced experts or special verification personnel to judge the situation, which cannot reflect the relative importance of each indicator and the subjective will is strong [26]. Entropy-TOPSIS method can objectively determine the index weight, according to the direct information value, the entropy-TOPSIS method effective combination, more scientific and reliable. Since the data of this paper come from the statistical Yearbook of Deyang City, the data are objective, meanwhile, it meets the requirements of the entropy-TOPSIS method, this paper uses this method to determine the weight of each system index. Details are as follows:

$$X_{ij} = \begin{cases} (x_{ij}\ \textit{is a positive indicator}) \frac{x_{ij} - \min\{x_{ij}\}}{\max\{x_{ij}\} - \min\{x_{ij}\}} \\ (x_{ij}\ \textit{is a negative indicator}) \frac{\max\{x_{ij}\} - x_{ij}}{\max\{x_{ij}\} - \min\{x_{ij}\}} \end{cases} \tag{4}$$

Calculate the weight value of j index in i year:

$$e_j = -\frac{1}{\mathrm{Inm}} \sum_{i=1}^{m} P_{ij} \mathrm{In} P_{ij} \tag{5}$$

The information entropy of each index is e_j:

$$e_j = -\frac{1}{\mathrm{Inm}} \sum_{\mathrm{i}=1}^{\mathrm{m}} \mathrm{P_{ij}} \mathrm{InP_{ij}} \tag{6}$$

In the formula, e_j is the information entropy of index j, P_{ij} is the proportion of j-th index of i-th evaluation object in the overall index,

Determining the weight of each index:

$$w_i = \frac{1 - e_j}{\sum_{j=1}^{n} (1 - e_j)}, \ \mathrm{j} = 1, \ldots\ldots, \mathrm{n} \tag{7}$$

$1 - e_j$ is the coefficient of variation. Afterwards, the positive and negative ideal solutions are determined:

$$Y^+ = (Y_1^+, Y_2^+, \ldots\ldots, Y_n^+) = (\max\{x_{i1}\}, \ldots\ldots, \max\{x_{in}\}) \tag{8}$$

$$Y^- = (Y_1^-, Y_2^-, \ldots\ldots, Y_n^-) = (\min\{x_{i1}\}, \ldots\ldots, \min\{x_{in}\}) \tag{9}$$

Calculating the Euclidean distance(L^+) from each sample to the ideal solution (Y^+) and the Euclidean distance(L^-) from each sample to the negative ideal solution (Y^-):

$$L_i^+ = \sqrt{\sum_{j=1}^{n} wj(X_{ij} - Y_j^+)^2} \tag{10}$$

$$L_i^- = \sqrt{\sum_{j=1}^{n} wj(X_{ij} - Y_j^-)^2} \tag{11}$$

Calculate the closeness of each evaluation object to the best solution S_i, to represent the development level of resilient cities and smart cities. The greater S_i is, the higher the development level of resilient cities and smart cities are.

$$S_i = \frac{L_i^-}{L_i^- + L_i^+} \tag{12}$$

Therefore, this paper will first use the entropy-TOPSIS method to calculate the weights of indicators, and then use the coupled coordination model for analysis. After that, in order to identify the main obstacle subsystems and the main factors of obstacles, an analysis of the main obstacle factors in the system will be conducted.

The degree of obstacle was calculated by the following formula, $\mathrm{F_j}$ is represented the weight of each index for the total target, h_j is the obstacle degree of each indicator based on the weighted sum for each year, while H_j is the obstacle degree value for each subsystem, the obstacle levels are divided to 5 levels, as is shown in Table 2.

$$\mathrm{D_j} = 1 - \mathrm{X_{ij}}, h_j = \mathrm{D_j} \times \mathrm{F_j} / \sum_{j=1}^{n} (\mathrm{D_j} \times \mathrm{F_j}) \tag{13}$$

$$\mathrm{H}_j = \sum h_{ij} \tag{14}$$

Table 2. Main Factor Obstacle Degree of Coupling Coordination for Resilience City and Smart City. (Self-drawn)

Proportion	Obstacle levels
[0, 10%)	Obstacle-free(0) or Low-obstacle level
[10%, 40%)	Mild-obstacle level
[40%, 60%)	Moderate-obstacle level
[60%, 90%)	High-obstacle level
[90%, 100%]	Extreme-obstacle level

2.3 Indicator System

This article, based on the ideas of systems theory, according to the current situation of Deyang City, divides the resilience city system into four subsystems: economic production, infrastructure, ecological environment, and resilience development. Combining the research and induction of resilience cities, it takes into account the systems of scholars such as Zhou Min [27, 28]. Based on the clusters of key terms, according to the original definition of resilience cities, as well as the requirements proposed by the Deyang Municipal Government, and referring to the resilience city construction requirements put forward in the overall land spatial planning of Deyang City, it aims to enhance disaster prevention, reduction, and public safety protection capabilities. It sets requirements for disaster assessment, early warning, monitoring, and rescue, as well as specific construction measures such as flood control, earthquake resistance, and emergency shelters. Ultimately, 18 indicators are selected. In terms of smart city construction, based on the requirements of the Deyang Municipal Government and the overall spatial territorial planning of Deyang City, a green, low-carbon, and smart municipal infrastructure support system is proposed, which demands the relationship between smart communication, smart management, and smart governance. Combining with related research on smart cities, the construction of the smart city system is divided into four subsystems: smart development, smart economy, intelligent facilities, and smart energy, from which 16 indicators are selected (Tables 3.1 and 3.2).

The specific indicators for this study are selected mainly from two aspects: the meaning of resilience and the specific impact of smart cities on people's livelihoods, that is, to promote the completion of urban disaster prevention and reduction as well as to ensure the disaster recovery capability, and to promote the interconnection of everything in the city. The specific indicators are combined with the sorting of keywords and previous research, and the data of the above indicators from 2019 to 2023 are finally selected for reference (each year's statistics reflect data from the whole previous year, so current data reflects the situation from 2018 to 2022).

From the perspective of survival, Economic Production is essential for daily food supply during disasters and post-disaster production and construction, so mainly considering the capabilities of agriculture and industry. At the same time, production capacity

Table 3.1. Resilience City System Indicators Overview Table. (Self-drawn)

Primary Indicators	Secondary Indicators	Tertiary Indicators	±
Resilient city system	Economic Production	GDP Growth Rate A_1	+
		Unemployment rate A_2	-
		Per Capita Output Value of Agriculture, Forestry, Animal Husbandry and Fishery A_3	+
		Per Capita Output Value of Industrial Employees A_4	+
		Total profits and taxes A_5	+
	Infrastructure	Water Bodies and Water Conservancy Facilities A_6	+
		Urban Per Capita Road Network Density A_7	+
		Highway Mileage A_8	+
		Comprehensive Production Capacity of High Water Supply A_9	+
		High Number of Beds Per Thousand Population in Medical and Health Institutions A_{10}	+
	Ecological Environment	Green Coverage Rate in High Built-up Areas A_{11}	+
		High Urban Sewage Treatment Rate A_{12}	+
		High Industrial Waste Emissions A_{13}	-
		Air Quality Compliance Rate A_{14}	+
		Carbon Emissions A_{15}	-
	Resilient development	Environmental Regulation Intensity A_{16}	+
		Number of Green Patents Obtained A_{17}	+
		Number of Judicial Cases A_{18}	-

is reflected in the overall economic volume and GDP growth, and a stable social environment requires a stable employment population. Therefore, the final selection of indicators includes A_1 to A_5.

The Infrastructure selected c. These indicators are related to urban stability and disaster response capabilities. For example, water is a necessity of life, and medical beds are important facilities for the recovery and treatment of the sick and injured. Transportation

Table 3.2. Smart City System Indicators Overview Table. (Self-drawn)

Primary Indicators	Secondary Indicators	Tertiary Indicators	±
Smart city system	Smart development	Smart Policy Strength B_1	+
		Government Technology Research and Development Funds B_2	+
		Number of R&D Personnel B_3	+
		Willingness to Learn through Intelligent Means B_4	+
	Smart economy	Number of R&D Activity Enterprises B_5	+
		Retail Revenue from Computers, Software, and Auxiliary Equipment B_6	+
		Willingness to Consume Online B_7	+
		Monthly Network Consumption Amount B_8	+
		Number of High-tech Enterprises B_9	+
	Smart Infrastructure	Number of Phone Users B_{10}	+
		Network Coverage B_{11}	+
		Electric Vehicle Ownership B_{12}	+

(*continued*)

Table 3.2. (*continued*)

Primary Indicators	Secondary Indicators	Tertiary Indicators	±
	Smart Energy	Energy Consumption Intensity B_{13}	-
		Coal Consumption B_{14}	-
		Industrial Production Consumes Electricity and Energy B_{15}	-
		Decrease Rate of Energy Consumption Per Unit of Regional GDP B_{16}	+

Note: + Positive Effects/ -Negative Effects

facilities and operations represent the ability to communicate and exchange resources with the outside world.

Ecological Environment indicators is concentrated on the area of ecological environment land and the annual pollutants, which will influence the environment.

Finally, Resilient Development views things that will stimulate the development of resilient cities, specifically from the occurrence of disasters, the stimulation of related policies, and the examination of innovation and social stability. Therefore, indicators as A_{16} to A_{18} was choosen.

Smart Development considers the impact of smart construction on the future, mainly the capacity for management services and talent innovation. Therefore, B_1 to B_4 are selected.

The Smart Economy takes into account the impact of intelligent construction on the economic industry, which includes the demand for intelligent construction from enterprises themselves, as well as the market consumer end, that is, the needs and changes of citizens towards a smart lifestyle. Therefore, certain indicators as B_6 to B_9 have been selected.

Smart Infrastructure is a basic requirement for the smart cities. In fact, Deyang City has already established smart transportation systems, smart government affairs, and smart education systems, which are difficult to evaluation quantification. Thus, indicators are considered from the perspective of smart transportation facilities and the mobile terminals needed for smart living, while the specific impact of the rest of the intelligent construction, such as the Internet of Things and information platforms, can be reflected in aspects like online consumer willingness, online consumption, smart policies, and technological output. Therefore, the final selection of indicators has been made by B_{10} to B_{12}.

Smart energy is an important part of intelligent construction. In conjunction with the current situation in Deyang, it is difficult to directly reflect the specific impact of intelligent construction, especially the data on the use of clean energy is not available. However, the smart energy system and smart energy production can be reflected indirectly through

indicators such as energy consumption intensity, coal consumption, industrial production electricity consumption, and the decline rate of energy consumption per unit of regional gross domestic product. Theoretically, the transformation of intelligent production can reduce pollution, and Deyang City is also promoting the construction of a smart ecological city. Therefore, indicators such as Energy Consumption Intensity are used to indirectly reflect the situation of smart energy, and the aforementioned indicators have been selected.

It is worth mentioning that, considering the requirements and characteristics of Deyang City's industrial city construction, as well as the difficulty in quantifying some of the results of smart city construction, specialized estimations have been made for a small portion of the missing data and a small portion of the data that are difficult to quantify. Explanations are provided as follows:

(1) A_7, calculating by query the data from the statistical yearbook by its definition
(2) A_{10}, calculating by query the data from the statistical yearbook by its definition
(3) A_{12}, data of 2019 is missing, estimated from the statistical method of linear regression.
(4) A_{13}, data of 2019 and 2021 is missing, estimated from the statistical method of linear regression.
(5) A_{15}, the data of which is obtained from the EDGAR (Global Atmospheric Research Emissions Database), which aims to work more closely together to collect and provide consistent CO2 emissions from fossil fuel combustion.
(6) A_{16}, according to the word frequency of Deyang municipal government work report, environmental protection, environment, ecology, nature and other words were selected to calculate their proportion.
(7) A_{17}, this data is confirmed based on the data obtained from the green invention patent retrieved by the China Research Data Service Platform.
(8) B_1, according to the word frequency of the work report of Deyang Municipal government, wisdom, science and technology, numbers and other words were selected to calculate their proportion.
(9) B_1, B_7, B_8, by the questionnaire survey, the content of different options was graded 1–5 and the score was obtained for calculation.
(10) The specific data of B_9 are supplemented according to the data on the website of the Sichuan Provincial People's Government and the 2019 Statistical Bulletin of The National Economic and Social Development of Deyang City, etc. However, the data in 2018 is missing and estimated by linear regression.
(11) B_{10}, using the number of Internet broadband access users divided by the total number of households in Deyang City.
(12) B_{11}, B_{12}, the data were obtained through the 14th Five-Year Plan of Electric vehicle charging infrastructure in Deyang City, and the missing data in 2022 was obtained through linear regression.

A_{13} (2019 and 2021) and A_{12} (2019) data were missing and had to be estimated using linear regression. This introduces uncertainty in the analysis for those years, However, relatively speaking, these two data use relatively scientific prediction methods, which have little impact on the overall evaluation. Besides, Data were collected from various sources, including government statistical yearbooks, environmental databases

(e.g., EDGAR), and policy documents etc. Different sources of information in different inclusion may lead to inconsistencies. The research used data from 2018 to 2022, which may not fully reflect the most recent developments, especially considering the dynamic nature of smart city and resilient city construction (Tables 4.1, 4.2, 4.3, 5.1, 5.2 and 5.3).

Finally, the relevant data are listed as follows, the units are removed, only the numerical values of the data are considered, and the data are obtained as follows:

Table 4.1. List of the index data values of the resilient city system. (Self-drawn)

Year	A_1	A_2	A_3	A_4	A_5	A_6
2018	8.9	3.7	10795	19.3	4408831	32356
2019	7.2	3.7	10569	15.5	3198314	32329
2020	2.5	3.5	12041	14.9	2932166	32329
2021	8.7	3.4	12456	19.4	4136496	29539
2022	3.1	3.2	13161	21.1	4911214	30732

Table 4.2. List of the index data values of the resilient city system. (Self-drawn)

Year	A_7	A_8	A_9	A_{10}	A_{11}	A_{12}
2018	6.4	8347	72.2	6.26634	41.1	91.4
2019	7.9	8627	72	6.85468	42.8	94.3
2020	9.2	10112	78.4	6.90001	43	95.4
2021	9.3	10379	75.6	7.12207	42.2	96.7
2022	9.6	10452	76.7	7.43011	43.5	97.7

Table 4.3. List of the index data values of the resilient city system. (Self-drawn)

Year	A_{13}	A_{14}	A_{15}	A_{16}	A_{17}	A_{18}
2018	19031	78.6	5205355	0.008876	274	4822
2019	17095	86.5	5068862	0.017009	221	6164
2020	15159	80.6	5210086	0.012044	336	5613
2021	8756	82.7	5471346	0.010490	355	5462
2022	6820	83.8	5379741	0.017085	351	5009

Table 5.1. List of the index data values of the smart city system. (Self-drawn)

Year	B_1	B_2	B_3	B_4	B_5	B_6
2018	0.01740	17565.4	10983	2.42	258	106249
2019	0.01494	17752	12820	2.53	355	58471
2020	0.01287	180342	13521	2.77	403	189292
2021	0.01341	244531	13407	2.85	390	263177
2022	0.01063	314157	15679	3.08	397	350576

Table 5.2. List of the index data values of the smart city system. (Self-drawn)

Year	B_7	B_8	B_9	B_{10}	B_{11}	B_{12}
2018	3.04	2.17	314	3997793	79.4	1202
2019	3.22	2.3	577	4394353	86.67	1698
2020	3.42	2.54	600	3913488	88.37	2714
2021	3.62	2.67	1070	3804339	159.5	6499
2022	3.85	2.89	1127	4197241	173.2	7255

Table 5.3. List of the index data values of the smart city system. (Self-drawn)

Year	B_{13}	B_{14}	B_{15}	B_{16}
2018	339.8	1530593	701305	4.9
2019	353.2	1457551	760947	3.1
2020	352.6	1329637	735747	0.8
2021	368.3	1196966	816420	1.5
2022	369.1	1181860	834674	4.1

3 Result

3.1 The Weight of Each Index and the Degree of Subsystem Obstacles

First, the weight calculation results in Tables 6 and 7 were obtained by the entropy-TOPSIS method. Among them, the weights of the four subsystems of Resilient City, Economic Production, Infrastructure, Ecological Environment and Resilient Development, are 29.48%, 25.62%, 28.73% and 16.17%, respectively. While for Smart City, the weights of Smart Development, Smart Economy, Smart Economy, Smart Infrastructure and Smart Energy are 23.90%, 26.40%, 24.40% and 25.30%, respectively.

Overall, the comprehensive evaluation of Deyang City's Resilient City construction and Smart City construction has shown a gradual upward trend. Among them, the comprehensive performance of Resilient City construction is quite impressive. From 2018

Table 6. List of index weight values of Resilient City system. (Self-drawn)

Index	A_1	A_2	A_3	A_4	A_5	A_6
Weight (%)	6.406	6.569	6.305	5.942	4.253	4.037
Index	A_7	A_8	A_9	A_{10}	A_{11}	A_{12}
Weight (%)	5.926	7.531	4.087	4.049	4.112	5.083
Index	A_{13}	A_{14}	A_{15}	A_{16}	A_{17}	A_{18}
Weight (%)	6.157	4.305	9.069	6.62	5.116	4.435

Table 7. List of index weight values of Smart City system. (Self-drawn)

Index	B_1	B_2	B_3	B_4	B_5		B_6
Weight (%)	4.99	8.525	4.583	5.812	3.716		6.025
Index	B_7	B_8	B_9	B_{10}	B_{11}	B_{12}	
Weight (%)	5.476	5.711	5.469	6.271	9.958	8.17	
Index	B_{13}	B_{14}	B_{15}	B_{16}			
Weight (%)	5.757	7.91	5.551	6.077			

to 2022, it rose from 0.299 to 0.833, with an overall increase of 0.534, and an average annual increase of 0.1335. The smallest increase was in 2019, with a growth rate of 24.38%, and the largest increase was in 2020, with a growth rate of 35.13%. It can be seen that the construction of a Resilient City is an increasingly positive trend, which is inseparable from the active investment and efforts of the Deyang municipal government in urban ecology and urban resilience construction. In contrast, the effectiveness of Smart City construction is slightly less than that of resilient city construction. Overall, from 2018 to 2022, over five years, the comprehensive evaluation index rose from 0.278 to 0.777, with an overall increase of 0.499 and an average annual increase of 0.1248. The fastest growth was in 2021, with a growth rate of 38.66%, and the slowest growth was in 2019, with a growth rate of 15.96%.

That is to say, both the two systems are getting better year by year. For the strong measures implementation by Deyang government, The construction of a green, low-carbon and intelligent municipal infrastructure support system in Deyang has been gradually improved in recent years. The construction of resilient cities and smart cities is a continuous process. Under the continuous policy encouragement and construction implementation, it will become more and more perfect, so its score should also be gradually increased. Since 2018, a series of strategies for ecological governance have been implemented, and the construction of smart production infrastructure has been started and continued. Smart production and ecological governance have gradually produced results, and the construction of urban security pattern is greatly conducive to the construction of resilient cities. From the overall upward trend, both straight lines tend to rise in a straight line, and their upward speed has a tendency to accelerate slowly. This

basically combines a series of strategies of Deyang for the development of ecological environment and smart city construction in recent years (Fig. 6).

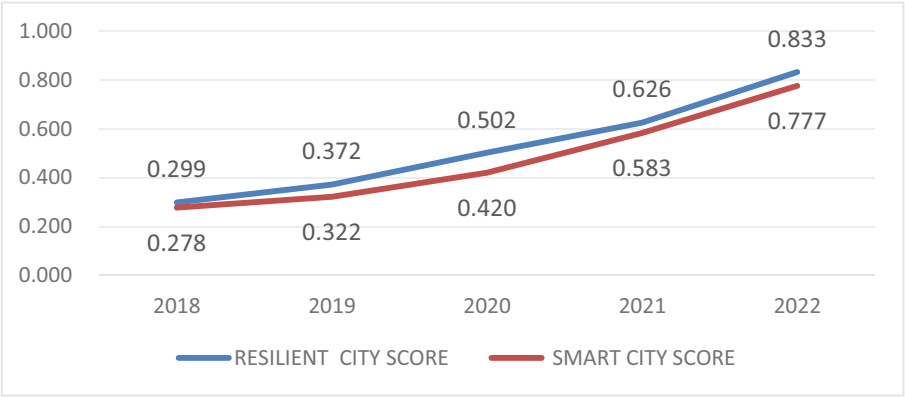


Fig. 6. Change chart of the comprehensive evaluation of resilient cities and smart city construction. (Self-drawn)

3.2 Coupling Coordination Degree Analysis and Obstacle Degree Analysis

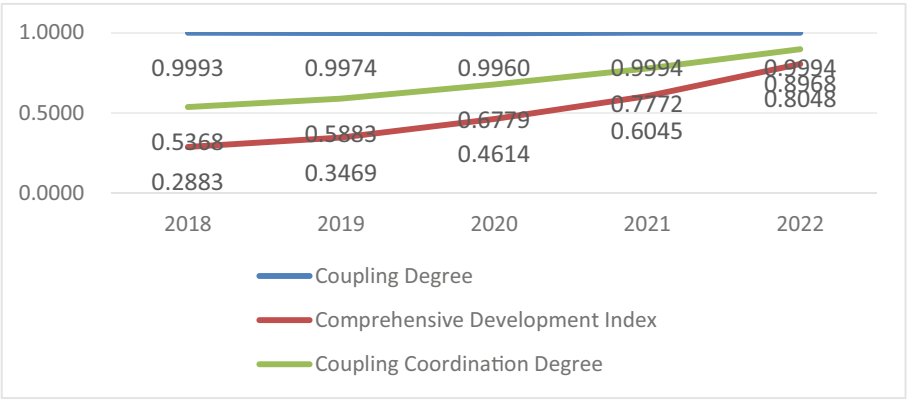


Fig. 7. Change map of the coupling and coordination degree of Resilient City and Smart City in Deyang. (Self-drawn)

According to the calculation of coupling coordination model, the coupling degree, comprehensive evaluation index, and coupling coordination degree curve of Deyang City's Resilient City and Smart City have been illustrated. Figure 7 shows that from 2018 to 2022, the coupling degree between Resilient City and Smart City in Deyang City has always been high, with the coupling degree basically above 0.9. It belongs to the Benign-trend. The annual change in the coupling degree between the two systems is small, all less

than 0.004, and the coupling degree continues to be higher than the coupling coordination degree. This indicates that there is a positive interactive and influential relationship between Deyang City's smart city construction and resilient city construction. Looking at the coupling coordination degree curve, its coupling coordination degree is in an accelerating upward trend, with the indicator data rising from 0.5368 to 0.8968. It can be seen that the coupling coordination is getting better year by year. Specifically, from 2018 to 2019, it was in the Mild-obstacle stage, with the overall levels at 0.5368 and 0.5883, respectively. Entering 2020 and 2021, the characteristics of coordinated development became more apparent, entering a good coordination relationship stage. Finally, in 2022, the dual systems achieved a High-quality-coordination development stage, ushering in a new era of development relations.

For the reason that Deyang have been highlighting the importance of urban resilience construction and smart construction. Therefore, Deyang has been trying to balance the two as much as possible when promoting their construction, ensuring that both aspects of the construction work are carried out at the same time, and can play a good effect. Therefore, the both systems showed a good coordination.

Scholars such as Luís M. A. Bettencourt [25] have repeatedly emphasized that cities are complex mega-systems, and theories related to management science also stress the need to consider from a systems thinking perspective. Therefore, the analysis of obstacle will start from the subsystem's obstacles and then consider the main obstacle factors. Due to the multitude of indicators, it is not possible to list and analyze each one. In the management process, managing the most critical methods according to the management philosophy of Pareto's Law can effectively enhance management effectiveness [26]. According to the principle of Pareto's Law, selecting the top 20% of 18 indicators and 16 indicators would be approximately 3 items. Therefore, in the optimization of this system's management, the main obstacle factors are also considered, and the top 3 most important items each year are selected for key analysis. It is hoped that governments and research teams can start with key issues to solve the dilemma of coordinated development between resilient cities and smart cities.

From the perspective of subsystem, the obstacles faced each year vary. As is shown in Tables 8 and 9, starting from the Resilient City, the main obstacles come from the in Economic Production and Ecological Environment, especially Economic Production which has greatly hindered the construction of Resilient City in 2019, 2020, and 2022. In 2020, the degree of hindrance exceeded 50%, reaching a Moderate-obstacle level and becoming a heavy factor restricting development. Subsequently, the obstacle degree of Ecological Environment for two years was slightly above 30%, belonging to a Mild-obstacle level. Similarly, the obstacles to the development of Smart City are also concentrated in two systems; the Smart Economy subsystem had the highest obstacle degree in 2018 and 2019, the Smart Energy subsystem in 2021 and 2022, and Smart Infrastructure only in 2020. Overall, from 2018 to 2020, the system's obstacle degree was relatively low, belonging to a Mild-obstacle level. However, in 2021 and 2022, it belonged to a High-obstacle level, which must be considered.

Analyzing from the main obstacle factors, shown in Tables 10 and 11, most of the which are at a Mild-obstacle level. For the Resilient City, A_2 and A_9 were main obstacle factors in the early two years, which have been controlled in the development of the

Table 8. List of obstacles to Resilient City subsystems. (Self-drawn)

Year	Economic Production	Infrastructure	Ecological Environment	Resilient Development
2018	26.25%	30.45%	30.81%①	12.49%
2019	44.85%①	26.86%	14.28%	14.00%
2020	54.13%①	6.76%	25.03%	14.08%
2021	25.85%	24.63%	30.63%①	18.88%
2022	34.78%①	26.67%	34.08%	4.47%

Table 9. List of obstacles to Smart City subsystems. (Self-drawn)

Year	Smart Development	Smart Economy	Smart Infrastructure	Smart Energy
2018	26.20%	35.18%①	30.94%	7.69%
2019	26.50%	29.34%①	24.62%	19.55%
2020	20.74%	21.66%	34.93%①	22.66%
2021	22.02%	13.95%	20.97%	43.05%①
2022	22.33%	0.69%	9.37%	67.61%①

Note: The highest obstacle subsystem of the year has been marked by ①

Table 10. A list of the main obstacle factors in Resilient City. (Self-drawn)

Year	RANK 1	RANK 2	RANK 3
2018	A_2/12.936%	A_9/10.406%	A_{13}/9.443%
2019	A_2/14.436%	A_9/11.988%	A_3/10.457%
2020	A_1/12.874%	A_4/12.672%	A_5/11.942%
2021	A_{15}/13.678%	A_{16}/13.226%	A_6/11.371%
2022	A_1/34.777%	A_{15}/23.671%	A_6/14.688%

following three years but still need attention. A_1, A_6, and A_{15} are the main obstacle factors in the latter three years, among which A_1 reached 34.777% in 2022, showing a trend towards a Moderate-obstacle level. The obstacle degrees of the two major factors, A_{15} and A_6, have also increased and should be given significant attention in the future. In terms of Smart City, the focus should first be on B_{11} and B_{13}, which are the main obstacle factors in the construction of Smart City, ranking first from 2018 to 2020 and from 2011 to 2022, respectively. B_2 and B_{12} were factors that occupied a relatively high position in the early years and have now been controlled, belonging to the Mild-obstacle level stage, and are expected to become Low-obstacle levels in the future, with moderate attention needed. Among the subsequent factors, B_{15} should be given special attention,

as it increased to 27.191% in 2022, belonging to the Mild-obstacle level stage, and the value is relatively high.

Table 11. A list of the main obstacle factors in Smart City (Self-drawn)

Year	RANK 1	RANK 2	RANK 3
2018	B_{11}/13.788%	B_2/11.803%	B_{12}/11.312%
2019	B_{11}/13.551%	B_2/12.567%	B_{12}/11.064%
2020	B_{11}/15.540%	B_{12}/10.576%	B_{16}/9.933%
2021	B_{13}/18.450%	B_{10}/15.037%	B_{15}/12.578%
2022	B_{13}/35.393%	B_{15}/27.191%	B_1/22.327%

4 Discussion

4.1 The Impact of the Annual Work on the Sub-System

Starting from 2018, Deyang City has made significant progress in ecological civilization construction. The city has focused on resolving environmental supervision issues, promoted supervision over environmental protection in accordance with the law, and promoted the treatment and rectification of black and odorous water bodies and county-level drinking water sources. At the same time, Deyang have started to integrate the development of smart city construction and ecological city construction. The pilot projects for soil pollution control and remediation are being accelerated, and the river (lake) chief system has been implemented effectively, with many key basin management projects carried out. These measures cater to the needs of resilient city construction, and Deyang City is also taking a path of ecological and smart development, closely integrating ecology and intelligence.

Therefore, for 2018, the Ecological Environment has a relatively large obstacle, as this year is the base year for this research, and these measures focus on the environment problems existing before. Because of these measures are just started, so it is surely to find the Ecological Environment to be the largest obstacle. This year's economic development has reached a new level, with the regional gross product exceeding 200 billion, and indicators such as fixed asset investment, fiscal revenue, and industrial added value have all achieved growth. The Chengdu-Deyang integration has achieved phased results, the Deyang-Mianyang integration has officially started. From the above content, it can explain why that the subsystems —Economic Production and Resilient Development— have a relatively low percent. This year's construction focus has rarely involved the Smart Economy field, so this year's Smart Economy subsystem is the most important obstacles.

In 2019, Deyang City faced a complex and severe situation. This year, the biggest obstacle sub-system of Resilient City is Economic Production, for the reason that influenced by the economic sanctions from the previous year that affected the overall domestic

economic environment. Besides, the advancement of "Deyang Smart Manufacturing" saw the development and deployment of a batch of "national important equipment," with the advanced manufacturing industry moving towards the middle and high end, that will make a great change in production, industrial production will enter a new adaptation period. Therefore, overall, the high obstacle degrees of Economic Production reflected the pangs of transformation, but also for the reason of the transformation, the data of Smart City subsystems performance very balanced, there are no overly prominent points. Smart Economy is the highest among the subsystems which is also a reflection of transformation.

In 2020 and 2021, the key targets were to maintain the stability of development because of transformation and the Covid-19 epidemic. In 2020, the economic structure and regional layout continued to be optimized, consumption continued to play a major driving role, with advanced manufacturing and modern services growing rapidly, ensuring the stability of Smart Economy and the resilience of Economic Production. Under the dual impact of the transformation and the epidemic, the obstacle degree of Economic Production is gradually close to 60%. During this year, the workers have a certain period of adaptation, in the process of promoting the smart transformation, the workers' education in the new mode of production and their familiarity with the new equipment will affect the work process, which directly affects Economic Production. At the same time, due to the overall impact of the epidemic, it is understandable that there are occasional quarantine policies and people-oriented production needs, but this will inevitably result in high Economic Production barriers. But Deyang City's new momentum continued to strengthen, with a number of significant scientific and technological achievements and the continuous growth of emerging industries, and the acceleration of traditional industry upgrades. Mass entrepreneurship and innovation were deeply carried out, with an average net increase of more than 10,000 enterprises per day, further promoting the development of Smart Economy. Due to the development of the Smart Economy and the need for smart production, the smart infrastructure has to keep pace, however, due to the epidemic and the lack of perfect infrastructure construction, the main obstacle to this year's Smart City system is from the Smart Infrastructure.

In 2021, under the continuous impact of the pandemic, Deyang City achieved new accomplishments in advanced manufacturing, with the Chengdu-Deyang high-end energy equipment industry cluster becoming one of the country's 25 advanced manufacturing clusters and newly selected as one of the top 100 advanced manufacturing cities in the country. That is to say, things get better in this year. After one year construction of smart infrastructure and the under control epidemic, production capacity has begun to become normal, and initial normal production has been able to take out under the epidemic. These have made a great contribution to the comprehensive evaluation of Economic Production and Smart Economy. It can be said that although the pandemic of Covid-19 affected production, it promoted the transformation towards smartization. Obstacle degree of the both subsystems decrease. In terms of ecology, the feedback from the central and provincial environmental protection supervision was orderly promoted, with the excellent day rate completing the provincial control target, It can be said that the ecological aspect is steadily advancing, so although the Ecological Environment has a high obstacle degree, it is only close to 30%. In addition, measures like 230 million yuan

in new entrepreneurship guarantee loans were issued, 384 cultural welfare performances were carried out, further enriching the spiritual and cultural life of the people, which contributes to the stability of Economic Production, Smart Economy, Smart Development, and Smart Infrastructure construction.

Finally, in terms of 2022, Deyang City has achieved significant success in both resilient city construction and smart city development. In terms of industrial transformation, with the mature of smart production, the performance of the Resilient City subsystems are quite well. Tianfu Digital Valley Digital Economy Industrial Park has attracted 105 enterprises, added 3,867 new 5G base stations, and become the first city in the province to build a gigabit optical network, with a total digital economy scale of 130 billion yuan, 53 new high-tech enterprises, 1 unicorn enterprise, 113 seed enterprises have been added, and the number of technology-based small and medium-sized enterprises has reached 850. All the things above show a good change in the field of smart construction, Smart Development,Smart Economy, and Smart Infrastructure are at a very good number. But this year, relatively little emphasis on smart energy development and construction, and that reflection at a very high number at sub-system Smart Energy.

4.2 Analysis on Achievements of Resilient City and Smart City Construction

The Development Promotion Mechanism is Stable and the Subsystem Operates Well. The talent and scientific and technological innovation, and green patents will all affect future construction efforts. Thanks to the active guidance and efficient development construction of functions for system has a significant impact on the outcomes of future construction. Factors sin the development process, showing no lag in future development. Relevant content should be steadily advanced, and in the future, policies for smart and green construction should continue to be promoted. Measures should be formulated to encourage these areuch as policy guidance, investment in promotion by the government, the subsystems resilient development and smart development have played a very positive role as, with substantial investment in scientific research to promote the steady development of green patents and R&D personnel. This will lead to social stability, relying on solid legal control to optimize judicial management and foster a positive social atmosphere to stabilize the implementation of relevant promotion policies.

The Construction of Urban Resilient Infrastructure and Smart Infrastructure has Been Gradually Improved. To promote the enhancement of urban resilience and improve the level of urban intelligence, the construction of resilient and smart infrastructure in cities must proceed steadily. Overall, the infrastructure advancement in this city is orderly, and full efforts have been made in public health to respond to the occurrence of the pandemic. The construction and completion of projects such as Deyang women and children's hospital, Deyang City People's Hospital's fifth-generation hospital in the north city, and the Deyang hospital affiliated with Chengdu University of Traditional Chinese Medicine have not only improved the level of medical facilities in Deyang city but also provided local residents with more efficient and higher-quality medical services. The capacity of roads, railways, and water supply has also gradually increased, which can produce effects on public emergency response, disaster prevention, and reduction. This has also shown by the change of *A9*, being one of the main obstacle factors in the first two years, after significant effects seen in sewage treatment and improvement of

supply ability, this factor has decreased in subsequent years. Smart infrastructure is also gradually well built, with urban smart facilities mainly controlled through the internet and mobile terminals, and medical care, education, and daily life have all been integrated into systems like WeChat mini-programs. by this way, the obstacle factors and degrees of smart infrastructure have remained at a low level, and the popularization of smart infrastructure is changing lifestyles while also promoting the improvement of infrastructure, which reflection by gradually decrease in obstacle percent number in smart infrastructure.

Strengthening the Gradual Shift Towards Resilient Production Methods to Replace Traditional Ones, and Perfecting Industrial Transformation with Intelligent Production Methods. In recent years, Deyang City has also been seeking a path of transformation and construction. The policy has highlighted the content of industrial transformation, and data also reflects some industrial upgrades and transformations have been completed. However, it must be said that the impact of the pandemic surely can be seen, coupled with the transformation of traditional industrial production, which has a deep impacted in the economic production before the Covid-19 pandemic arrived in 2019. The sanctions that began in 2018 also made the economic situation in 2019 and 2020 difficult. The main obstacle factor A_2 precisely illustrates how transformation have affected the employment rate. For instance, the local Sichuan College of Architectural Technology, which is primarily focused on the construction industry, due to employment saturation, transformation and upgrading are urgently needed to improve the employment rate. Of course, the effectiveness of policies has a lag, so after the normalization of the pandemic in 2021, the economic obstacles began to decrease, and transformation gradually started to show results. The obstacle degree of the Smart Economy system is gradually decreasing, but to promote Smart Economy, the pain of transformation, economic sanctions, and the chain reactions caused by the pandemic are real. Therefore, the improvement in Economic Production is low, and with Resilient Development, Smart Development policies, and the effects of Smart Economy on Economic Production requiring time and exploration, the overall obstacle for Economic Production is relatively large, which is also reflected in A_1 becoming the main obstacle factor. Obstacle factors A_2 and A_1 should be given close attention to ensure they do not break through or approach the moderate obstacle level; otherwise, it would result in a very poor effect, just like the situation when the pandemic hit in 2020. In addition, A_3, A_4, and A_5 also have certain obstacles, although not significant, they also need to be vigilant. Although the tasks related to rural revitalization have been carried out and implemented, the future productivity of rural areas and the transformation of agricultural production also need to be considered. This requires future efforts to better strengthen economic transformation work, promote policy implementation, and enhance the promotion of intelligent production methods.

The Effect of Ecological Construction is Stable, and the Transformation of Energy Production Needs to be Further Strengthened. To ensure the ecology, it is necessary to promote greening construction and promote urban green development, using ecology to promote civilization. Especially as an ecological greening city, Deyang must closely integrate smart methods to promote ecological construction. Green space level, air quality, pollution level, and carbon consumption level are all important indicators for

monitoring urban ecology. Although Deyang has always attached great importance to ecological construction, on the one hand, due to being an industrial city, the industrial transformation has not been fully completed, so Ecological Environment may still be an obstacle. This relies on the industrial transformation to reduce the consumption of A_{15}. On the other hand, the upgrading and transformation of industries will promote the improvement of the ecological environment, which also makes Ecological Environment at a slight obstacle level in recent years. Although the pandemic and the construction process of industrial transformation will still bring some negative impacts, the obstacle degree of Ecological Environment is still relatively controllable. The effect of Smart Energy needs further control, as the proportion of clean energy is immeasurable, and its power generation situation can only be measured from data indicators such as coal consumption. Therefore, it can be said that the obstacles of B_{13} and B_{15} are still quite high. Therefore, to promote clean energy and ensure B_{13} and B_{15}, this will also promote the reduction of A_{15}.

4.3 Suggestions for Further Work

Overall, the construction of Smart City and Resilient City promotes and coordinates with each other, and has currently reached a level of High-quality-coordination development. Therefore, this study is mainly based on the analysis of the main obstacle factors in coupling coordination, and combined with work in recent years, the following suggestions for future development are proposed.

Actively Promoting Reformation of University Education. First, strengthening school-enterprise cooperation to build a platform for integrating education with industry. Encouraging universities to cooperate with local enterprises in Deyang to jointly build laboratories, internship bases, etc., providing more practical opportunities for students and enhancing their employability. Second, focusing on optimizing curriculum settings and focusing on the cultivation of applied talents. Adjusting the curriculum structure, increasing the proportion of practical courses, and cultivating students' ability to solve practical problems, better adapting to the needs of industrial development. For example, the proportion of AI courses and language ability courses is added to the course, so as to promote students to become modern talents who can adapt to the development of international and intelligent development. Third, implementing innovation and entrepreneurship education to unleash students' potential. Encouraging students to engage in innovation and entrepreneurship activities, and offering policy support and financial aid when feasible, to nurture their innovative spirit and entrepreneurial awareness, thereby broadening their capabilities and providing a more robust competitive edge. Teachers can teach students to use building reconstruction, or base practice, jointly invest and contract with the government, use short video platforms such as TikTok and Kuaishou, or use wechat circle of friends, spread channels, establish their own marketing model and team, and start a business independently. Fourth, to establish and improve an employment guidance service system. For example, universities such as the Sichuan College of Architectural Technology provide guidance on employment information, career planning, job-seeking skills, etc., which can better help students find employment smoothly, especially in industries like construction where the job market may be challenging.

Promoting Industrial Transformation Steadily and Increase Investment in Industrial Research to Ensure the Quality and Upgrading of Industries. First, government must spare no effort to promote the transformation of industry, encourage enterprises to adopt new technologies and processes, enhance the added value of products and market competitiveness, and promote the development of traditional industries towards high-end and intelligent directions. Leaders at all levels and those responsible must take the initiative to learn relevant knowledge and implement clear applications and incentive policies for new technologies and processes, so as to effectively promote this transformation. For example, unmanned aerial vehicles, robotic arms and other technologies are applied to the production process to improve the efficiency and safety of production. Second, strategic emerging industries must be cultivated, increasing support for strategic emerging industries such as new materials, new energy, and biomedicine, and create new economic growth points. Finally, strengthening scientific and technological innovation to enhance the competitiveness of industries. Establishing and improving the scientific and technological innovation system, encouraging enterprises to increase R&D investment, enhance independent innovation capabilities. The government can provide some promotion policies, scientific research scores, and incentive funds to improve this content.

Gradually Improve the Local Energy Consumption Structure and Promote the Application of New Energy Products and Production. To optimize the energy structure and reduce the percentage of coal consumption, it is necessary to advance the substitution of clean energy and increase the utilization ratio of renewable energy sources. By introducing mature wind power production or solar power production technology from the power system or improving the equipment and equipment of the electric power production sector can do it. In addition, the new energy vehicle industry should be promoted, with increased efforts in the promotion and application of new energy vehicles and the improvement of supporting facilities. Strengthening the construction of electric vehicle infrastructure, enhancing the perfection of charging piles and automatic control systems, plan advanced methods such as autonomous taxis for the future, and promoting the transformation of personnel towards automotive maintenance services. Finally, advocating the use of energy-saving technologies and supporting the adoption of energy-saving products, improving energy utilization efficiency, and reduce energy consumption.

Unswervingly Promoting the Ecological and Smart Cities Construction, and Gradually Promoting the Intelligentization of Ecological Management Facilities. First, an ecological monitoring network should be built to enhance ecological management capabilities. Promoting the construction of smart ecological monitoring systems to monitor the quality of the ecological environment in real-time, and promptly identify and resolve environmental issues. Utilizing advanced IoT technologies to address water system issues and continue to promote the intelligent management of black and odorous water bodies. While employing new methods, this can also stimulate new employment opportunities. Second, promote green buildings and low-carbon transportation, completing the S11 rail transit as soon as possible, energy saving and carbon reduction will be achieved through the construction of public transport infrastructure. In addition,

strengthening ecological protection and restoration. Continuously enhancing ecological conservation and restoration initiatives, while developing ecological parks, wetland parks, etc., to improve the quality of the urban ecological environment, provide more ecological activity spaces for citizens, and increase the ways citizens can engage in ecological leisure. Finally, promoting the integrated development of ecological industries, develop ecological tourism, ecological agriculture, and other ecological industries to achieve a win-win situation for both economic and ecological benefits.

5 Conclusion

Overall, Deyang did a good job in handling the relationship between Smart City and Resilient City, the comprehensive evaluation and coordination relationship between both are gradually rising. In the future, we should continue to pay attention to important obstacles, solve major problems, and do a good job in resilient development and smart development to ensure the work of transformation. The index system and evaluation method of this study are of reference value to similar studies, but each city has its own characteristics and ways of development, the specific evaluation method should be adapted to local conditions and evaluated scientifically.

Acknowledgments. This study is supported by the project Analysis and research on the coupling nature of smart city and resilient city construction in Deyang City from the perspective of city integration circle, from Deyang City Federation of Social Sciences, Project No. DY25ZX053.

References

1. Cheng, J., Chen, R.: A study on the coupling and coordination between urban resilience and regional economic development level in the Yangtze River Delta urban agglomeration. J. Jinling Sci. Technol. Soc. Sci. Ed. **38**(03), 18–25 (2024). https://doi.org/10.16515/j.cnki.32-1745/c.2024.03.003
2. Xia, Z., Ren, F., Wang, Y.: A study on the relationship between urban infrastructure ecological transformation and urban resilience from the perspective of complex systems. China Environ. Manag. **16**(04), 108–117 (2024). https://doi.org/10.16868/j.cnki.1674-6252.2024.04.108
3. Xia, C., Zhai, G.: A study on the development path of comprehensive disaster prevention planning under the territorial spatial planning system. Shanghai Urban Plan. **2**, 67–73 (2024)
4. Khatibi, H., Wilkinson, S., Dianat, H., Baghersad, M., Ghaedi, K., Javanmardi, A.: Indicators bank for smart and resilient cities: design of excellence. Built Environ. Project Asset Manag. **12**(1), 5–19 (2022). https://doi.org/10.1108/BEPAM-07-2020-0122
5. Nozarian, M., Fereidunian, A., Hajizadeh, A., Shahinzadeh, H.: Exploring social capital in situation-aware and energy hub-based smart cities: towards a pandemic-resilient city. Energies **16**(18), 6479 (2023). https://doi.org/10.3390/en16186479
6. Voghera, A., Giudice, B.: Evaluating and planning green infrastructure: a strategic perspective for sustainability and resilience. Sustainability **11**(10), 2726 (2019)
7. Liao, H.: Research on the optimization strategy of urban overall security resilience under the background of "digital intelligence". Nanjing Univ. Posts Telecommun. (2023). https://doi.org/10.27251/d.cnki.gnjdc.2023.000483

8. Xie, X., Chen, J., Liu, N.: A study on the evaluation of urban resilience: a case study of Deyang City. Ind. Saf. Environ. Protect. **46**(10), 77–81 (2020)
9. Huang, H.: Research on the development strategy of smart city construction under the background of urban integration: a case study of Deyang City. Construct. Econ. **40**(05), 103–107 (2019). https://doi.org/10.14181/j.cnki.1002-851x.201905103
10. Yang, Y.: Deyang civilian pass smart city new media platform: empowering people's livelihood with digitalization. China Broadcast. Televis. **21**, 53–56 (2022)
11. Li, G., Wang, R.: A study on the coupled and coordinated development of regional tourism economy and ecological environment in Guangxi. J. Dali Univ. **9**(07), 48–57 (2024)
12. Li, Z., Liu, X., Liu, F.: Optimization strategies for public spaces in old communities based on resilience theory: a case study of Quanshui pavilion street community in Jinan. Hous. Sci. **42**(10), 38–42 (2022). https://doi.org/10.13626/j.cnki.hs.2022.10.001
13. Qi, X., Luan, B., Zhou, W., et al.: A study on the influencing factors of urban coastal ecological resilience based on the STIRPAT model: a case study of Shenzhen city. Chin. J. Environ. Sci. **45**(1), 1–14 (2024)
14. Huang, J., Hang, B., Wang, F., et al.: Application of vehicular network technology in the construction of smart transportation in Xiangyang. Smart Cities **10**(01), 20–23 (2024). https://doi.org/10.19301/j.cnki.zncs.2024.01.006
15. He, B., Guo, R., Zhang, C., et al.: Methods for constructing natural scenes for digital twin cities. Surv, Map. Bull. **07**, 87–92 (2022). https://doi.org/10.13474/j.cnki.11-2246.2022.0209
16. Xi, G., Zhen, F., Xiang, X., et al.: Research progress and prospects on the matching of urban life service supply and demand under the influence of intelligent technology. Prog. Geograph. **42**(11), 2231–2241 (2023)
17. Han, S., Wang, H., Ao, Y., Wang, B., Chen, B., Martek, I.: Resilient city construction efficiency and its influencing factors in China's Chengdu-Chongqing economic circle: considering both construction input and resilience level of the city. Sustain. Cities Soc. **114**, 105726 (2024). https://doi.org/10.1016/j.scs.2024.105726
18. Fu, K., Liu, Y., Lu, X., Chen, B., Chen, Y.: Health impacts of climate resilient city development: evidence from China. Sustain. Cities Soc. **116**, 105914 (2024). https://doi.org/10.1016/j.scs.2024.105914
19. Makhoul, N., Navarro, C., Sung Lee, J.: Seismic estimation of casualties and direct economic loss to Byblos city: a contribution to the '100 resilient cities' strategy. Sustain. Resilient Infrastruct. **7**(3), 201–221 (2022). https://doi.org/10.1080/23789689.2020.1745531
20. Wu, M., Guo, M., Xu, J.: The influence of smart city policy on urban green energy efficiency – a quasi-natural experiment based on 196 cities. J. Cleaner Prod. **449**, 141818 (2024). https://doi.org/10.1016/j.jclepro.2024.141818
21. Cho, K., Park, H., Kim, C., Kim, T.: Toward sustainable urban energy solutions: Introducing F NeCESSCity framework for energy-saving smart city configuration. Sustain. Cities Soc. **113**, 105674(2024). 10. 1016/j.scs.2024.105674
22. Kumar, A., Akhtar, A.K., Pandey, A.: Design of internet of things system based smart city model on Raspberry Pi. IETE J. Res. **69**(12), 8781–8788 (2023). https://doi.org/10.1080/03772063.2022.2088629
23. Zhou, M., Yang, L., Ye, D.: Study on the coupling and coordination of smart city construction and green development in Guiyang city. Land Nat. Res. Res. **05**, 34–39(2024) .https://doi.org/10.16202/j.cnki.tnrs.2024.05.007
24. Miao, C., Liu, L., Zhang, A.: Measurement, obstacle factors and coupling analysis of green finance and high-quality economic development level in the Yellow River Basin. J. Anhui Univ. Sci. Technol. Soc. Sci. Ed. **26**(04), 13–22 (2024)
25. Zhang, L., Liu, R., Zheng, D., et al.: Relationship between advanced manufacturing agglomeration and regional economic–innovative–social coupling coordination development: a spatial analysis in Liangjiang new area, China. J. Urban Plan. Dev. **150**(2) (2024)

26. Muhadaisi, A., Fei, Z., Weng, N.C., et al.: Corrigendum to coupling coordination analysis and spatio-temporal heterogeneity between urbanization and eco-environment along the silk road economic belt in China. Ecol. Indic. **121**, 107014 (2021)
27. Ogundunmade, P.T., Adepoju, A.A.: Modelling E-commerce data using Pareto principle. Mod. Econ. Manag. **3**, 11 (2024)
28. ArunKumar, M.S., Ganesh, N.M., et al.: Optimization of process parameters in micro electrochemical machining using TOPSIS technique with analytical hierarchy process (AHP). Eng. Res. Express **6**(3), 035534 (2024)
29. Zhou, Z., Hu, Q., Zhu, Q., et al.: Research on urban resilience measure based on integrated modeling of neural networks. J. Wuhan Univ. Technol. Inf. Manag. Eng. Ed. **43**(06), 505–511 (2021)
30. Zhang, A., Miao, C., Chen, Z.: Analysis of urban ecological resilience, social network and its influencing factors in the Yellow River Basin. Geograph. Arid Region **48**(01), 130–142 (2025)